

# Improving Feature Extraction from Construction Drawings

PRINTO — AI Drawing-to-ERP Platform

Coral Business Solutions | RealSoft ERP | June 2026 | Phase 2

## 1. Executive Summary

Printo reads construction drawings and extracts ~26 structured fields (title block, room schedule, dimensions, materials), validates them against 18 rules, and pushes them to RealSoft ERP. This document explains **how we improve extraction accuracy** and which improvements are **now integrated** into the pipeline.

Two architectural decisions shape this phase: (1) all AI inference is routed through a **pluggable AI sidecar** (the Pratyaya / Ampris AI-aaS pattern) rather than calling a model API directly from Printo; and (2) every improvement is made **measurable** by a field-level benchmark harness so gains are provable, not assumed.

## 2. Current Pipeline & Gaps Addressed

The original pipeline sent the whole file to a vision model with a free-text JSON prompt and trusted the model's self-reported confidence. Three concrete gaps were found and fixed:

| Gap                                                       | Impact                                         | Fix                                          |
|-----------------------------------------------------------|------------------------------------------------|----------------------------------------------|
| Deterministic prepass disabled (pdfplumber not installed) | Free, reliable title-block text was never used | Dependency added; prepass active + expanded  |
| quantities validated (R10/R18) but never extracted        | Rules always warned; field always blank        | Added to schema, prompt, and prepass regex   |
| Raw self-reported confidence                              | Unreliable; confidence rules (R12–R15) noisy   | Replaced with a calibrated score (Section 5) |

### 3. The Improvement Levers

Fifteen techniques, grouped by where they plug into the pipeline. The Tier column marks what is integrated now (**Tier 1**, **Tier 2**) versus the roadmap (**Tier 3**, Section 7).

#### A. Give the model a better image

| # | Technique                                               | Why it helps                                                  | Tier |
|---|---------------------------------------------------------|---------------------------------------------------------------|------|
| 1 | Render PDF pages to high-DPI PNG                        | Small title-block text becomes legible vs. a raw PDF stream   | 1    |
| 2 | Title-block crop pass (zoomed bottom / bottom-right)    | Title blocks are tiny corners; a zoomed crop reads far better | 1    |
| 3 | Image cleanup: deskew, CLAHE contrast, upscale, denoise | Rescues scans and phone photos of drawings                    | 2    |
| 4 | Tiling large sheets into regions                        | Beats resolution limits for dense counts / schedules          | 2    |

#### B. Strengthen the deterministic prepass (free, reliable)

| # | Technique                                   | Why it helps                                         | Tier |
|---|---------------------------------------------|------------------------------------------------------|------|
| 5 | PDF text-layer extraction + expanded regex  | Exact title-block values with no model cost or error | 1    |
| 6 | OCR fallback (Tesseract) for scanned sheets | Recovers text when there is no PDF text layer        | 2    |

#### C. Use the model better

| # | Technique                                        | Why it helps                                       | Tier |
|---|--------------------------------------------------|----------------------------------------------------|------|
| 7 | Structured output + schema validation (Pydantic) | No JSON-parse failures; typed, repaired, retryable | 1    |

|    |                                                   |                                                |   |
|----|---------------------------------------------------|------------------------------------------------|---|
| 8  | Multi-pass (title-block pass + full-plan pass)    | Each pass specialises; merged for best of both | 2 |
| 9  | Few-shot / golden title-block templates           | Anchors the model to known layouts             | 3 |
| 10 | Drawing-type routing (plan / section / elevation) | Type-specific field sets and prompts           | 3 |

#### D. Reliability & trustworthy confidence

| #  | Technique                                       | Why it helps                                       | Tier |
|----|-------------------------------------------------|----------------------------------------------------|------|
| 11 | Confidence calibration (multi-signal)           | Confidence reflects reality, driving rules R12–R15 | 2    |
| 12 | Self-consistency voting (run N×, reconcile)     | Stabilises uncertain fields                        | 3    |
| 13 | Cross-sheet aggregation of project-level fields | Project / client agree across a sheet set          | 3    |

#### E. Close the loop & prove it

| #  | Technique                                                 | Why it helps                                   | Tier |
|----|-----------------------------------------------------------|------------------------------------------------|------|
| 14 | Human-in-the-loop from the <code>corrections</code> table | Real corrections become examples / tuning data | 3    |
| 15 | Benchmark harness (field-level accuracy)                  | Every change is measured before/after          | 1    |

## 4. What We Integrated (Tier 1 + Tier 2)

| Capability                                              | Module / file                      |
|---------------------------------------------------------|------------------------------------|
| PDF→PNG render, title-block crop, image cleanup, tiling | <code>backend/preprocess.py</code> |

|                                                       |                                                          |
|-------------------------------------------------------|----------------------------------------------------------|
| Text-layer + OCR prepass, expanded regex, quantities  | backend/prepass.py                                       |
| Structured output, schema validation & repair         | backend/schema.py                                        |
| Pluggable AI provider (mock / sidecar; vision & text) | backend/ai_provider.py                                   |
| Orchestration, multi-pass, confidence calibration     | backend/extractor.py                                     |
| Accuracy benchmark + ground truth                     | backend/benchmark.py,<br>test_drawings/ground_truth.json |
| Local test sidecar (contract reference)               | tools/mock_sidecar.py                                    |

## 5. Confidence Calibration

Self-reported confidence is unreliable, so Printo computes a calibrated per-field score by combining independent signals:

- **Prepass agreement** — a deterministic text-layer/OCR match sets the field to near-certain (0.97).
- **Format validity** — drawing-number pattern, revision pattern, 1:NN scale: small boost if valid, penalty if not.
- **Cross-field consistency** — e.g. `sheet_number ≤ total_sheets`; violations lower both scores.
- **Coverage floor** — every populated field gets a sensible minimum so the UI always shows a score.

## 6. Sidecar Architecture

Printo never embeds a model key. `ai_provider.py` routes inference to a local sidecar (`SIDECAR_URL`, default `http://localhost:8787`) and falls back to built-in mock data when the sidecar is unreachable (`AI_PROVIDER=auto`). It supports a **vision** mode (sends the rendered image) and a **text** mode (sends OCR text), auto-selected from the sidecar's advertised capability.

| Endpoint         | Contract                                                                                                             |
|------------------|----------------------------------------------------------------------------------------------------------------------|
| GET /health      | <pre>{ "status":"ok", "vision": true false,   "model": "..."} </pre>                                                 |
| POST /v1/extract | <pre>body { prompt, schema,   image?{media_type,data(b64)}, text? } → {   "data": { fields..., confidence } } </pre> |

**Swap-in ready:** point `SIDECAR_URL` at the real Pratyaya AI-aaS (or any service implementing this contract) for genuine vision extraction — no Printo code change required. `tools/mock_sidecar.py` documents and stubs the contract.

## 7. Benchmark Methodology & Roadmap

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`benchmark.py` runs the pipeline over the sample drawings and scores each field against `ground_truth.json` (booleans by equality, lists by overlap, strings normalised with substring tolerance), printing per-file and overall accuracy. Run it before and after a change, or across providers, to quantify impact. Example: the mock provider scores 100% on the filename-aware samples; a fixed stub sidecar scores ~81%, correctly flagging the floor mismatches — demonstrating the harness detects real differences.

**Tier-3 roadmap (next phase):** golden-template matching, drawing-type routing, self-consistency voting, cross-sheet aggregation, and a human-in-the-loop feedback loop fed by the existing `corrections` table. Real accuracy figures require pointing the sidecar at a live vision model.